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How artificial intelligence can enable data classification for market sizing - Insights from applications in practice

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ABSTRACT

Determining the size of the addressable market is a key aspect of market intelligence and requires identifying and delineating projected budget data from potential customers. The market intelligence arena is characterized by a wide range of disparate sources, many of which are unstructured, ranging across competitive, market, financial, and technology sources, and typically necessitating significant manual work to analyze, reconcile, and integrate. The authors present an approach for classification of data from one of these sources, facilitating aggregation and analysis of intelligence information. We describe a concept proof using machine learning that extends a model for automatic mapping of publicly available budget data to segments and subsegments of a market segmentation taxonomy. This approach automates the tagging of market and market segment for each program and cost element by training classification models on the manually labeled historical data. We describe the evaluation and use of multiple natural language processing (NLP) and classification modeling methods. This work's contribution is demonstrating how NLP and machine learning techniques can provide useful data classification and automatic classification even when source data diverges from its specified taxonomic description.

1. Introduction

Market intelligence involves information gathering and synthesis across multiple domain Kelley (1965), is an iterative process (Jaworski et al., 2002), and requires both data integration and classification with a unified view of multiple, inconsistent data sources (Lenzerini, 2002). Developing a market strategy for new products requires both market intelligence and competitive intelligence, processes that are both continuous and complementary (Jamil, 2013). Competitive intelligence ranges across business model, market, customers, product, technology, and human resources (Pellisier & Nenzhelele, 2013), is integral to business strategy (Maritz & Du Toit, 2018), improves market agility (Tarn & Wang, 2023), and is positively related to organizational performance (Yin, 2018) (Tahmasebifard, 2018). One core aspect of competitive intelligence is gaining knowledge about the size and composition of the addressable market. Market size for a potential product is a critical factor that enterprises must know before choosing to invest in developing that product.

The rapid increase in available market information is a potentially significant benefit to enterprises. Recent advances in machine learning (ML) techniques have created an opportunity to accelerate a largely manual market intelligence data integration and reconciliation process (Bharadiya, 2023). However, applying machine learning requires both careful selection of the most suitable ML techniques and an analysis of the contextual challenges such as poor-quality data and inconsistent existing data models. Considerable care must be exercised when applying data analytics to achieve organizational outcomes, balancing both quantitative and qualitative analyses (Allen, 2023). Responsible artificial intelligence requires considerations that includes technical and social components that must be balanced (Akbarighatar et al., 2023). Artificial intelligence algorithms have been used to improve market segmentation (Chang & Fan, 2023), and to select market segments to maximize overall profit (Peng et al., 2022). Successful segmenting contributes to profit by better understanding customers and markets (Rosario & Dias, 2023). Supervised algorithms were used in Peng et al. (2022) to build a market segmentation model. Our work considers semi-supervised and supervised algorithms, but ultimately uses supervised classification algorithms in conjunction with natural language processing (NLP) embeddings models. Market intelligence data is often collected from external sources and tends to contain unstructured data types (Jaworski et al., 2002), predominately free-form text descriptions. These unstructured data types are converted into vector representations,

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prior to building the classification model, using embeddings models. We implemented Gradient Boosting, a supervised machine learning ensemble model for the classification of market segments. While Gradient Boosting is not a deep learning model, we've found that deep learning models do not always perform better depending on the underlying data structure and resources available (Schmitt, 2023).

This paper describes the application of ML techniques to determine market segment sizing in an effective monopsony, a market dominated by a single buyer, applying machine learning to the process of tagging budget data from the single buyer to determine the size of each market segment. By consistently tagging each datum of projected budget information, sorting it into an appropriate group, and then analyzing each group of related data, we can provide business guidance on the expected size of the market. Our objective was to find an effective set of ML techniques to greatly reduce the time required to develop business insights on projected market and market segment sizes from available budget data. ML techniques have been applied to multiple aspects of firm size and performance in economics in Saha et al. (2023), but were not used to project market size. More generally, ML techniques have been used for demand estimation in economics, associated more with product-level measures than market size measures, although product aggregations can act as a proxy for market size (Bajari et al., 2015).

We describe:

- The challenges of data collection and preparation
- The problems encountered with an existing manually constructed taxonomy
- Our application of text embedding techniques
- Our application of semi-supervised classification techniques
- Our application of automatic market segmentation classification techniques

Following sections present our results with a discussion of the observed merits of each technique. We review the challenges encountered toward meeting our goal of automated market segment sizing within the defense industry and describe the ML approaches implemented in practice toward achieving this goal. We conclude with a discussion of the implications of our work and suggestions for future research directions.

2. Background and related work

2.1. Motivation

The central challenge for our team was to reduce the time and cost associated with categorizing budget data. The existing process required substantial effort from highly trained analysts. We chose to focus on a part of the heavily manual current process: the task of categorizing budget data from published customer organizations into markets and market segments. The resulting data can then be used for decision support. The current process relies on an internally developed taxonomy, and we were provided examples of past analyses with manually tagged budget data.

Disparate data must be extracted from sources, cleaned, contextualized, and then sorted ("tagged") into useful domain categories usually represented as a taxonomy. The categories represent market segments, and the construction of a taxonomy can be a subjective process (Tonks, 2009). Segmentation categories can be derived from market and customer and require consistency in their unit of analysis (Sausen et al., 2005). Whether done manually, as in the past, or with automation support, this classification process is vital. Fig. 1. shows this flow.

Each source of data has a particular context that provides the meaning and significance of the data. The specific meanings of the terms employed in the data depend highly on context (Haase, 2004). While tagging is common, the use of well-structured taxonomies is less so. At present, the most precise way to capture the meaning of data is in formal

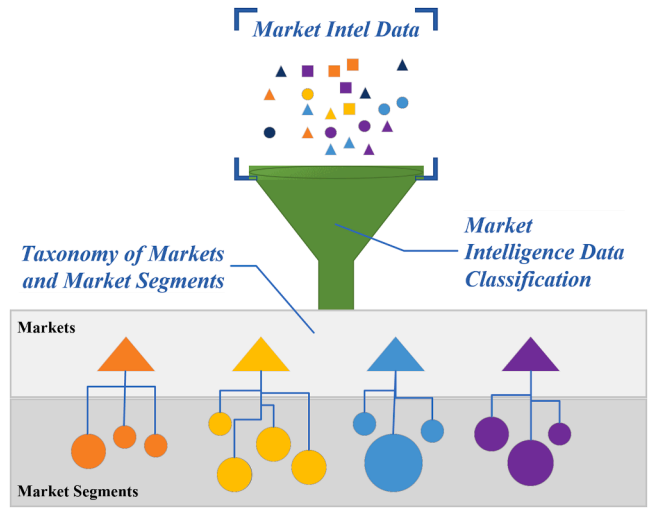

Fig. 1. Classification sorts market data into a taxonomy.

representation of meaning using semantic structures and richly relational ontologies (Grimm, 2009), as "the often-invisible structures of metadata profoundly impact the meaning that can be derived from that data" (Canning et al., 2022). The current practice is a laborious manual data tagging, and the lack of a solid ontological structure has limited automated approaches.

2.2. Problem formulation

We hypothesized that we could effectively tag the data by using an existing rough taxonomy as a unified set of workable "bins" or "categories." The problem would then become a classification one: how best to classify each of the data elements to the most appropriate place in that taxonomy. While some approaches further address categorization of taxonomical elements with separate machine learning algorithms (Thorat et al., 2021), in our work we only needed to go one level deeper in the taxonomy. For that stage of our process, we developed a classification approach to build satisficing models for the most critical slices of the data. We built models for grouping budget information into "markets" and "market segments" using historical data for training. These models were trained on historical data that was manually tagged by Subject Matter Experts (SMEs), and the resulting models provided the needed data classification for competitive intelligence.

At the project's onset, we had anticipated that a previously used taxonomy associated with our data sources could directly be used to seed the classification groups. We discovered substantial inconsistency between the data and the taxonomy that it was using. We suspect this is far more common in practice than the ideal case of finding accurately tagged data. Our approach allowed us to overcome this potential barrier. The Natural Language Processing (NLP) based approach described herein provides a bridge to overcome the lack of rich accurate semantic data descriptions. One goal was to speed up the delivery of tagged and curated budget data to our business stakeholders. Our premise was that by automating the market and market segment tagging for each cost element using classification models trained on manually labeled historical data, we could reduce delivery time and increase the quality of the tagged and curated budget data we deliver.

The manually tagged DoD (Department of Defense) budget data also contained a sizable number of unlabeled records (80,734 records out of 98,570 records were unlabeled, approximately 82 %). Labeling such data is a very time consuming and an expensive process (Settles & Craven, 2008).

3. Methodology

We used common performance measures to determine the success of the classification models and decide on the viability of automated tagging. These included confusion matrices (Fig. 2 Market Model Confusion Matrix and Fig. 3 Market Segment Model Confusion Matrix below), F1 scores, precision, and recall values based on a test set (Hossin & Sulaiman, 2015). The weighted F1 scores, mean of the per-class F1 scores weighted by the number of occurrences of the class in the dataset, were used to select between models. Berente et al. (Berente et al., 2019) describe a "computationally intensive" approach to theory development that aligns with the experimental and iterative nature of our methodological investigation, while Kar et al. (Kar et al., 2023) have detailed both specific steps and pitfalls to avoid (Kar et al., 2023).

Before selecting the precision and recall values of each class were considered to ensure the selected model could both find the relevant classes in the test dataset and identify only the relevant data points. The precision and recall values can be used to identify if a class imbalance problem exists in the dataset. The F1 score is a combination of precision and recall and is used as a measure to balance between the two.

3.1. Data collection

The Market Intelligence team historically collected DoD budget data and manually tagged the budget line items with market and market segment tags based on Raytheon's market intelligence taxonomy. These tags were used to analyze the addressable market for our aerospace business unit. We collected three years of domestic budget data covering the period from 2019 to 2021. This data contained 31 customers and 2505 program elements. The data contained a mixture of categorical and unstructured text fields. Subject Matter Experts (SMEs) spent a significant amount of time labeling the data each quarter when a new DoD budget was released.

3.2. Data preparation

An initial data quality assessment was conducted to evaluate the project's feasibility. Some data quality issues related to inconsistent naming conventions, low occurrence values, duplicate records, and missing data were discovered. We chose to focus on our single primary source, DoD customer budget data (29). With appropriate data cleaning, we believed we could extract useful business insights. Fields that contained greater than 50 % of the data omitted or were not common between each year's budget extract were removed. Cramer's V (Cramér, 1999) and Factorized Pearson Correlations (Hall, 2000) were used to identify variables containing redundant information. Variables that were highly correlated with another were removed prior to modelling. Data dictionaries were used to recode data values for handling inconsistencies. These inconsistencies included misspellings and using abbreviations or acronyms in some cases while spelling out the full word or words in others.

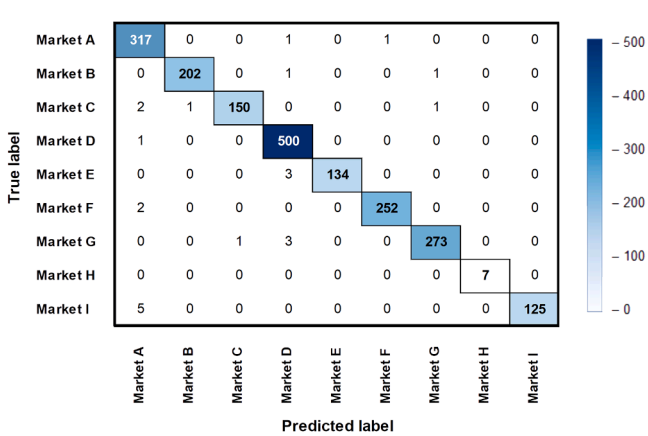

Fig. 2. Market model confusion matrix.

To train our models, it was necessary to clean the text prior to creating vector representations. This cleaning activity included (1) filtering out stop words, common words that add little value, such as 'the', 'is', 'a', (2) lemmatizing strings to return the root word, (3) removing punctuation, numbers, and special characters, and (4) making all characters lower case.

3.3. Text embedding and feature selection

The unstructured and semi-structured text columns were converted into numeric vector representations prior to creating a classification model. The categorical variables may be one-hot-encoded, but the unstructured text required a Natural Language Processing (NLP) embedding model. One-hot-encoding is the process of assigning columns a one or zero value based on the presence or absence of a category. With unstructured data, there was too high of a cardinality of features to encode in this manner (Cerda & Varoquaux, 2020). The result was a large, sparse matrix with a new column for each record in the dataset.

Multiple text embedding models were developed, applied, and tested on each of the seven fields requiring an NLP model (in Tables 3 and 4). These included utilizing pre-trained models and applying a transfer learning approach as well as building models trained on the budget data.

The pre-trained models included utilizing Python's SpaCy library and GloVe model, en_core_web_lg, to create word embeddings (Pennington et al., 2014). An average vector was calculated from the word embeddings associated with each record. Another pre-trained model, Google's Universal Sentence Encoder, was also applied to create document embeddings (Cer et al., 2018). This model was trained at the sentence level and uses a DAN (Deep Averaging Network) (Iyyer et al., 2015). The last pre-built encoder attempted was from the Transformers library and utilized the Hugging Face model, all-mpnet-base-v2, to map sentences and paragraphs to a 768-dimensional dense vector space (Devlin et al., 2018). While the pre-trained embeddings performed well on some of the variables, they were inadequate at capturing the particular language used in the application domain of others.

Term frequency inverse document frequency (TfIdf) vector representations were attempted, but created vectors that were too large and sparse (Harish & Revanasiddappa, 2017). Term frequency counts the number of times a word was used in a document and the inverse document frequency is used to down-weight words that have a high global frequency, meaning they are common across all documents. Since the TfIdf approach created sparse vectors, a Latent Semantic Analysis (LSA) model was built to resolve the problem (Deerwester et al., 1990). LSA uses Singular Value Decomposition (SVD) to reduce the dimensions while creating a semantic representation. While the LSA model improved upon the TfIdf approach, it was ultimately not used in the final classification models as other architectures performed better. Word2Vec (Mikolov et al., 2013) and Doc2Vec (Le & Mikolov, 2014) embedding models were created for the variables that did not perform adequately using the pre-trained transfer learning approach. Word2Vec was used on fields that contained only a few words, while Doc2Vec was used when the field contained a paragraph of data per a record. The Word2Vec models are based on a continuous skip-gram model that predicts words within a certain range before and after the current word. An average of the word vectors is used to compute the record or row vector. Doc2Vec can be seen as an extension of Word2Vec and computes a feature vector for every document (row) instead of each word. When training the models, a small vector size was used with more epochs to accommodate our smaller corpus. The window, maximum distance between the current and predicted word, was set to three and the learning rate and minimum learning rate were kept at the defaults of 0.025 and 0.0001 respectively.

The text embedding vectors, and one-hot-encoded variables were

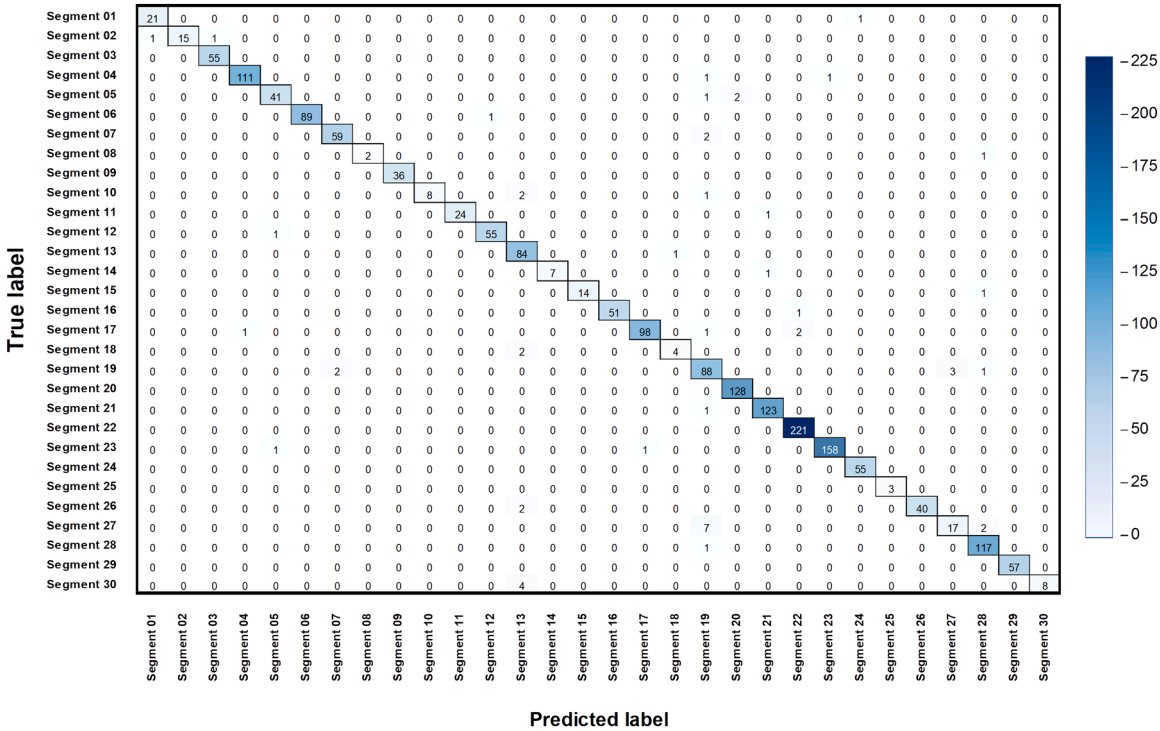

Fig. 3. Market segment model confusion matrix.

combined into a single data frame that was then split into labeled and unlabeled data sets. The labeled data frame was then split into the features and target variable. Feature selection techniques were used to reduce the feature space. The feature selection techniques included ANOVA (Analysis of Variance) (Freedman, 2009), LASSO (Least Absolute Shrinkage and Selection Operator) (Tibshirani, 1996), and Random Forest Classifier. PCA (Principal Component Analysis) (Abdi & Williams, 2010) was also tried as a dimensionality reduction approach to limit the size of the feature space, however, feature selection performed better than PCA on the test set. The down selected features and target variable was further split into a training and test set with an 80/20 split.

4. Experimentation

4.1. Semi-supervised classification

On the surface this situation appeared to be a prime candidate for semi-supervised learning, because we had some labeled data and a large amount of unlabeled historical data. Semi-supervised learning or weak supervision is a type of training that combines labeled with unlabeled data to make predictions on new examples (van Engelen & Hoos, 2022). It can be used as an approach to increase the volume of your training data.

In addition to training a classification model on the training set of labeled data, the authors also applied semi-supervised modelling approaches within the classification model constructs using a combination of the labeled and unlabeled budget data from 2019 to 2021. For each variable, different models and techniques might give better results. Instead of relying on a single approach, the authors used the choice of classification technique itself within our classification model. Rather than perform an attempt for the full combinatorial set of possible configurations, we made assumptions based on the known performance of the techniques to get a manageable set of attempts. As we progressed, the performance of specific techniques allowed us to further restrict the set of possible attempts to those most likely to yield good results.

A semi-supervised training dataset was created that combined a subset of the x-variables from the labeled training set with the x-variables from the unlabeled dataset. A 'no label' dummy variable was concatenated with the training y-variable entries. The semi-supervised approaches tried included Label Propagation (Zhu & Ghahramani, 2002), Label Spreading (Zhou et al., 2003), and Self-Training (Nigam & Ghani, 2000). A K-Nearest Neighbors (KNN) kernel with 7 neighbors, 10,000 max iterations and a convergence tolerance of 1E-2 was applied to both the Label Propagation and Label Spreading experiments. The nodes assign labels based on neighbors to infer clusters.

The semi-supervised models were trained on the mixed data set of labeled and unlabeled training data to return assigned labels for the entire mixed set. These assigned y labels were then used with the x-variables of labeled and unlabeled data to train the classification models. This process was repeated for all attempted classification models for both Label Propagation and Label Spreading techniques. The test set weighted F1 scores, in conjunction with the class precision and recall scores, were used to evaluate the performance of the classification models using Label Propagation and Label Spreading versus the classification model alone.

A Self-Training approach was also attempted for the Random Forest and Gradient Boosting classification models. The Self-Training worked by iterating over high-probability pseudo-labels. We fit a model on the labeled training set and predicted the unlabeled data set. A class probability is returned with the predicted label. High probability predictions or pseudo-labels were added to the training data and removed from the unlabeled data set. The model was then re-trained with the original labeled data and the high probability records. The re-trained model would then be used to predict the remaining unlabeled data. This process of re-training and adding high probability predictions to the training set was repeated until there were no more high probability predictions.

The outcome of our experiments with semi-supervised classification models (Label Propagation, Label Spreading, and Self-Training) found that contrary to our initial thoughts the classification models alone performed better on the test set. We posit that the semi-supervised models did not perform as well because the labeled records were not randomly assigned. This could have implications on applying the current classification models to all new DoD quarterly budget entries. The

data may need to be initially filtered to remove items that are different from the labeled data set the models were trained on.

4.2. Classification

The authors built multiple classification models (Support Vector Machine, Logistic Regression, Random Forest Classifier, and Gradient Boosting) to evaluate the performance of these pre-processing techniques such as text encodings, feature selection, and dimensionality reduction, and tried various combinations of the above-mentioned techniques. A classification model would be built using a text embedding technique for each text variable and a feature selection or dimensionality reduction technique on the encoded dataset. The performance measures for each trial were recorded for comparison and ultimately the selection of the best performing classification model and feature engineering approach applied.

As with the semi-supervised classification approach, we needed to create a set of combined techniques for each classification model. Once again, as we progressed, the performance of specific techniques allowed us to further restrict the set of possible attempts. The classification models tried were Support Vector Machine (SVM) (Vapnik, 1998) , Logistic Regression (Wright, 1995), Random Forest (Breiman, 2001), and Gradient Boosting (Friedman, 2001). Model parameters were tuned using a cross validation grid search scored by weighted F1 score. Various combinations of text embedding models for each of the unstructured text fields and each of the feature selection techniques were tested against each of the listed classification models.

The compute resources required, and the time allocated towards each trial are limitations of the experimentation approach. To evaluate each combination's performance measures, a trial run must be executed against the training data for each combination and evaluated against the test set. To address these evaluation challenges, not all possible combinations were tried, and a focused subset of combinations were executed. Subject matter expertise, prior knowledge of approaches that work best together, and a comparison of the possible modelling approaches in relation to our data set were used to focus the experimentation.

The best performing classification model based on weighted F1 score was Gradient Boosting using LASSO selected features for both market and market segment classification models. The parameter settings were selected by cross validation grid search using a weighted F1 score for each of the classification models and the selected parameters are presented below in Table 1 and Table 2.

Table 3 and Table 4 list the selected text embedding techniques applied to each of the free-form text variables. The output of these are the encodings that were used in LASSO feature selection and to build the chosen market and market segment classification models.

5. Results

The chosen model parameters, the text embedding models used on each text variable, the weighted and class performance measures, and the confusion plots are provided in the tables and figures below. The displayed text embedding models used for each text variable with LASSO selected features, and the model parameter settings below when used to tune a Gradient Boosting classification model, produces the best tried model as measured by weighted F1 score. Overall, the selected market and market segment classification models performed fittingly on the test datasets. We observed a 99 % accuracy for the market classification model and a 97 % accuracy for the market segment model. The performance indicators were higher than initially anticipated and deemed adequate for proceeding with automated tagging.

The parameter settings applied to the chosen Gradient Boosting model are presented below in Table 1 and Table 2. After the feature engineering, text encoding and feature selection is complete, a Gradient Boosting model is tuned using a grid search and the below parameters were selected based on the weighted F1 score. The market and market segment parameters below result in the highest F1 score for the selected combination of text embedding models, feature selection, and classification model.

The chosen text embedding models and their associated number of dimensions, prior to feature selection, for each target model can be seen in Table 3 and

Table 4 below. These embedding models were chosen in relation to the selected classification model as a combination trial set. Another combination may be more performant when applied to an alternate classification model.

The summary evaluation metrics found for the selected models are

Table 1
Best parameters for market model.

Parameter	Value
Subsample	1
N Estimators	1000
Min Samples Split	6
Min Samples Leaf	3
Max Features	Sqrt
Max Depth	4
Learning Rate	0.05

Table 2
Best parameters for market segment model.

Parameter	Value
Subsample	0.95
N Estimators	1000
Min Samples Split	6
Min Samples Leaf	13
Max Features	Sqrt
Max Depth	3
Learning Rate	0.05

Table 3
Unstructured text embedding models: Market.

Field	Text Embedding	Number Dimensions
Cost Element	Word2Vec	40
Program Element Name	SpaCy (en_core_web_lg)	300
Raytheon System Name	Word2Vec	40
Raytheon Subsegment	DAN (Google Sentence Encoder)	512
Program Description	Doc2Vec	40
Platform Name	FastText	40
Subcontractor	Word2Vec	40

Table 4
Unstructured text embedding models: Market segment.

Field	Text Embedding	Number Dimensions
Cost Element	Word2Vec	40
Program Element Name	SpaCy (en_core_web_lg)	300
Raytheon System Name	Word2Vec	40
Raytheon Subsegment	DAN (Google Sentence Encoder)	512
Program Description	Doc2Vec	80
Platform Name	FastText	40
Subcontractor	Word2Vec	40

Table 5
Market classification model scoring.

Metric	Value
Accuracy	0.988
Weighted F1	0.988
Weighted Precision	0.989
Weighted Recall	0.988

shown in Table 5 and Table 6. These include the overall accuracy and weighted evaluation measures. Both the market and market segment classification models performed satisfactorily in aggregate. The market classification model has a weighted F1 score of 0.99 and the market segment model has a weighted F1 score of 0.97. Each of the evaluation measure values are rather uniform, indicating we have a balanced model.

The class evaluation metrics are presented below in Table 7 and Table 8. The market classification model performed well on each class. All class F1, precision, and recall scores are above 0.95. The market segment classification model overall performed well; however, a few classes have less stellar evaluation metrics. The reason for this is the increase in the number of classes and the available occurrences of each class. We have thirty classes to predict amongst and some of those classes had very few samples to train against. As expected, the classes with few samples performed better in precision than recall.

The Market Model and Market Model Segment Confusion Matrices are plotted below in Fig. 2 and Fig. 3, respectively. The output from the models was rendered in Python, exported as contingency tables, and formatted into the figures below using Microsoft Excel for readability.

The market and market segment classification models performed well against the test set and were deemed trustworthy for automated tagging of the DoD budget data.

The next step was to deploy the models, so they could be used by the Market Intelligence team to automatically label the quarterly budget. The Python scripts were integrated into an R Shiny application using the reticulate library to source the Python scripts. The Shiny application was used as a dashboard front-end for running inference on the market and market segment classification models and performing all of the necessary pre-processing, text embedding, and feature selection. The user may utilize the application to upload the latest DoD budget and have the tool display the data with the automatically tagged market and market segments via a user interface.

Market Intelligence Analysts upload the latest market budget data on a quarterly basis. The stored models and the market and market segment labels are added to the uploaded data generating inferences to support strategy development.

6. Discussion

6.1. Implications to practice

We began with the hope that an existing proprietary taxonomy would serve as a guide to classification of budget data for markets and market segments. The primary business goal was to support market strategy development with market intelligence. Previously this function was entirely a manual process. A key benefit is the reduced labor costs realized. It is no longer necessary to manually label each DoD budget line item. As an additional benefit, we also saved the time and expense spent training analysts while improving labeling consistency.

Finding discrepancies between the pre-tagged historical data and that taxonomy alerted us to deeper challenges. We chose to focus on the most proximate, classifying single-source data from available inputs. The goal must always be to aid the user, to derive insights from the data. Our work met that immediate need, but also exposed a second "deeper challenge" — how to best characterize and capture the context for a given data source.

Table 6
Market segment classification model scoring.

Metric	Value
Accuracy	0.972
Weighted F1	0.971
Weighted Precision	0.972
Weighted Recall	0.972

Table 7
Market model evaluation metrics.

Variable Name	F1 Score	Precision	Recall
Market A	0.981	0.969	0.994
Market B	0.993	0.995	0.99
Market C	0.984	0.993	0.974
Market D	0.991	0.984	0.998
Market E	0.989	1	0.978
Market F	0.994	0.996	0.992
Market G	0.989	0.993	0.986
Market H	1	1	1
Market I	0.98	1	0.962

Table 8
Market segment model evaluation metrics.

Variable Name	F1 Score	Precision	Recall
Segment 01	0.955	0.955	0.955
Segment 02	0.938	1	0.882
Segment 03	0.991	0.982	1
Segment 04	0.987	0.991	0.982
Segment 05	0.943	0.953	0.932
Segment 06	0.994	1	0.989
Segment 07	0.967	0.967	0.967
Segment 08	0.8	1	0.667
Segment 09	1	1	1
Segment 10	0.842	1	0.727
Segment 11	0.98	1	0.96
Segment 12	0.982	0.982	0.982
Segment 13	0.939	0.894	0.988
Segment 14	0.933	1	0.875
Segment 15	0.966	1	0.933
Segment 16	0.99	1	0.981
Segment 17	0.975	0.99	0.961
Segment 18	0.727	0.8	0.667
Segment 19	0.893	0.854	0.936
Segment 20	0.992	0.985	1
Segment 21	0.988	0.984	0.992
Segment 22	0.993	0.987	1
Segment 23	0.991	0.994	0.988
Segment 24	0.991	0.982	1
Segment 25	1	1	1
Segment 26	0.976	1	0.952
Segment 27	0.739	0.85	0.654
Segment 28	0.975	0.96	0.992
Segment 29	1	1	1
Segment 30	0.8	1	0.667

This paper describes a single data source used for integration, but multiple sources are likely to be needed to best support the business needs. Data in the wild can be elusive, misrepresented, and disorganized. It is necessary to better understand the context of each datum to correctly capture its meaning and implications. Our examination of both the existing taxonomy and many of the taxonomies associated with other potential sources suggests something more is needed to correctly handle context characterization and cross-context data integration. To address the challenge of human error in data generation, the challenge of poorly defined data meaning, and the challenge of highly varied data contexts, market and competitive intelligence may require semantic techniques which embody reified human experience complemented by Natural Language Processing and Machine Learning techniques.

6.2. Contributions to literature

This work has shown how a mélange of NLP and machine learning techniques can provide useful data classification and automatic classification even when source data diverges from its specified taxonomic description. The distance between the hidden semantics inherent in the data and the specified taxonomy associated with the data is strongly similar to the distances between the various semantic structures (whether implicit or richly formal) that come with ingesting a wide

variety of data sources.

This parallel suggests the success of our core NLP and classification models when strengthened by an integrated ontology and associated semantic integration processes may provide the basis for a robust competitive intelligence data integration framework. Many taxonomies have been proposed for market information and the broader context of competitive intelligence. Osterwalder's BMO (Business Model Ontology) ranges across customers, capabilities, revenue, and costs, and facilitates analysis of a firm's competitive strategy (Osterwalder, 2004;Osterwalder et al., 2005). Samavi, et al. describe a conceptual framework for strategic reasoning about business models that extends Osterwalder's work to include a strategy layer, separating operational and strategic concerns and directly considering intent of the various actors in a competitive environment (Samavi et al., 2009). A causal path between strategy and market outcomes has been described ontologically for competitive intelligence that includes market segments (De Rozario, 2009), and business model elements are compared for strengths and weaknesses.

A layered set of ontologies can be assembled from established sources with specific additions for the need of the application at hand. See Fig. 4 for an example of such an approach. We believe additional work would show this technique of semantic engineering is a powerful knowledge representation structure that can ground our derived models. This technique could support the "meaning integration" necessary to both correctly identify source contexts for data and to bridge disparate contexts, gleaning insights from a wide variety of data sources.

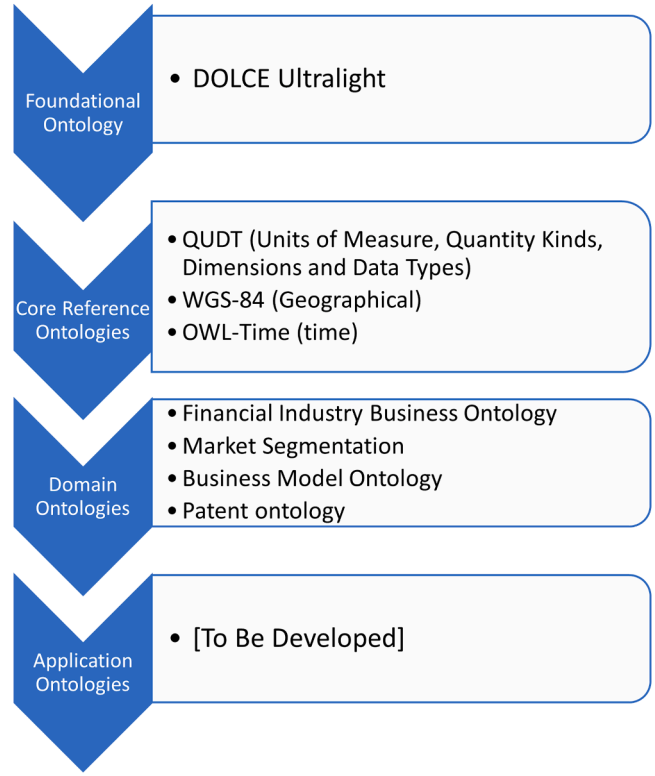

Fig. 4. A layered ontology to support future source integration.

7. Conclusion

To develop business strategy, one must establish the size of the addressable market. This requires market intelligence which in turn must draw on a wide variety of data sources. In the past, one aspect of this process was to manually tag publicly available budget data to establish the size of markets and market segments. In our attempt to automate this process, we discovered the expected difficulties of unstructured data, but also the unexpected challenge of incorrect use of a provided taxonomy in historic data. This single issue highlighted a key weakness in the subjectivity of taxonomic approaches. A degree of subjectivity is inevitable, but when a taxonomy becomes subjectively idiosyncratic to the point where it fails to work well with the data, it must be set aside.

We were able to circumvent the taxonomic problem using a collection of classification techniques that built a series of trained models to successfully tag the budget data for our prime source. This met the minimum requirements for the immediate business need, and we have provided details of the evaluation for each approach we employed. Significantly, our work shows that taxonomic approaches that we had hoped to use for additional sources were unlikely to provide a clear path to integrating data across those multiple sources. Regardless of how thoroughly additional sources are tagged, adhering to their individual taxonomic structures, the semantic distances between the respective data context becomes a barrier to effective integration. Instead of supporting deeper understanding, the added sources of data would be likely to confound and confuse the business picture. In our view the path forward will lie in a combination of machine learning classification techniques with an ontologically supported semantic integration approach to extract and preserve the meaning in the data.

CRediT authorship contribution statement

L. Stallings: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **P. Bhat:** Writing – review & editing, Project administration, Data curation, Conceptualization. **J. Jacobs:** Writing – review & editing, Writing – original draft, Visualization, Conceptualization. **K. Lynch:** Writing – review & editing, Writing – original draft, Validation, Supervision, Project administration, Conceptualization. **Q. Risch:** Writing – review & editing, Validation, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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